

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO4: Articulation (BL4, BL5)	Assessment:	Assessment Tool 3 – Comparative Analysis
Weightage:	10% of CIA	Total Marks:	100
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
Student Name:	AKSHAYA KUMAR CHANDRASEKAR	Reg. No.:	192425350
Section / Batch:	B.TECH AIML	Date:	22-08-2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze the given questions thoroughly before answering.
- Compare multiple aspects such as methods, results, advantages, and limitations clearly.
- Critically evaluate the strengths, weaknesses, and implications with proper justification.
- Present meaningful insights, interpretations, and well-supported conclusions from the comparisons.

Question Type	Focus Area	Questions
Comparative Analysis	Comparative analysis of Dynamic Programming and Greedy algorithms for optimization problems.	Q1

SECTION A – Comparative Analysis

ASSESSMENT CO4-AT3-COMPARITIVE ANALYSIS

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Language:

Python

C

C++

Java

QUESTIONS

Q1

Matrix Chain Multiplication (DP vs Greedy)

100 marks

1/1 answered

Q1 Matrix Chain Multiplication (DP vs Greedy)

100 marks

Given matrices with dimensions (10×20), (20×30), (30×40), compare Dynamic Programming with a naive greedy approach for multiplication order. Analyze cost differences and correctness. Evaluate why greedy fails. Derive insights on optimization strategies.

Your Code

```
1 def matrix_chain_order_dp(p):
2
3     n = len(p) - 1
4     m = [[0] * (n + 1) for _ in range(n + 1)]
5     s = [[0] * (n + 1) for _ in range(n + 1)]
6
7     for L in range(2, n + 1):
8         for i in range(1, n - L + 2):
9             j = i + L - 1
10            m[i][j] = float('inf')
11            for k in range(i, j):
12
13                q = m[i][k] + m[k + 1][j] + p[i - 1] * p[k] * p[j]
14                if q < m[i][j]:
```

Input

stdin input

Output

Given matrices: A1(10x20),
A2(20x30), A3(30x40)

Dynamic Programming (Optimal)
Approach:
Optimal parenthesization : ((A1

Code of Conduct

I **AKSHAYA KUMAR CHANDRASEKAR** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: **AKSHAYA KUMAR CHANDRASEKAR**

RUBRICS FOR CALCULATION

Comparative Analysis					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%				
Comparison & Differentiation	25%				
Critical Evaluation	25%				
Synthesis & Insight Generation	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____